

4.9GHz–6.1GHz 3W Transmitter Module (VTX)

User Manual

V1.0

2026. 05



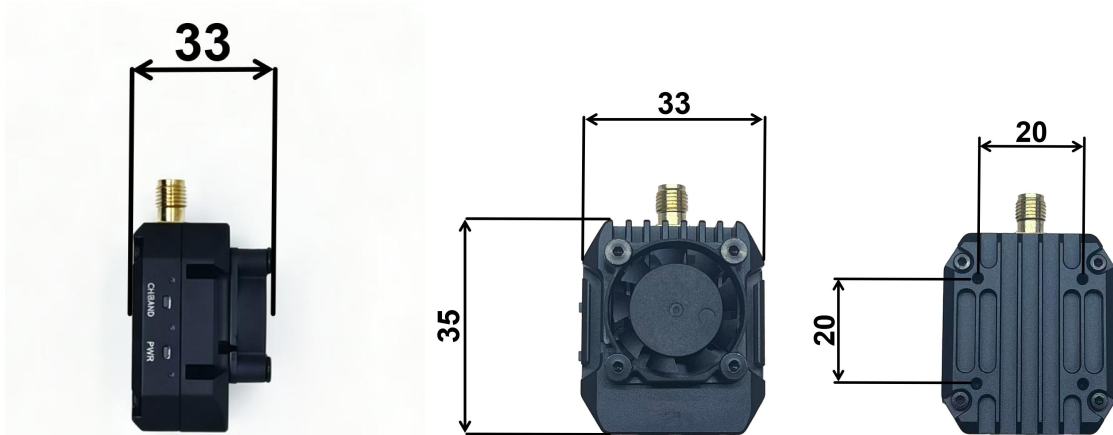
1.1 Part Name



1. SH1.0-8P cable
2. Channel Indicator (Red LED)
3. Band Indicator (Blue LED)
4. Power Indicator (Green LED)
5. Channel/ Band(CH/BAND) Button
6. Power Setting (PWR) Button
7. Antenna

1.2 Dimensions

The module dimensions and mounting dimensions are shown below (unit: mm); please reserve adequate installation clearance.



1.3 Specifications

- a. 64 channels: 4.9 GHz – 6.1 GHz (4950 MHz – 6050 MHz)
- b. Max. output power: 10 W (adjustable: 25 mW / 1 W / 2 W / 3 W)
- c. Supply voltage: DC 8 V – 36 V, 2 S–8 S Li-ion/Li-Po compatible
- d. Antenna connector: SMA female
- e. Control protocol: IRC Tramp
- f. Cooling: aluminum enclosure with heatsink and cooling fan
- g. Mounting pattern: 20 mm × 20 mm, $\varnothing$ 2.0 mm holes
- h. Weight: 56 g

2.1 Usage Instructions

2.1.1 Setting the Channel

Short-press the CH/BAND button to change the channel. Each press advances to the next channel; the corresponding red LED blinks once to indicate the selection (CH 1 → 2 → 3 → 4 → 5 → 6 → 7 → 8, then wraps back to CH 1). See the diagram below for details.



Short-press the CH/BAND button to switch channels								
	CH1	CH2	CH3	CH4	CH5	CH6	CH7	CH8
RED LED	Blink once	Blinks twice	Blinks three times	Blinks four times	Blinks five times	Blinks six times	Blinks seven times	Blinks eight times

2.1.2 Setting the Band

Long-press the CH/BAND button to switch bands. Hold for 5 s: the blue LED blinks to indicate the selected band, cycling through groups A → B → E → F → R → P → H → U. Refer to the diagram below for details.



Long-press the CH/BAND button for 5s to switch bands								
	Band A	Band B	Band E	Band F	Band R	Band P	Band H	Band U
Blue LED	Blink once	Blinks twice	Blinks three times	Blinks four times	Blinks five times	Blinks six times	Blinks seven times	Blinks eight times

2.1.3 Setting Transmit Power

Press the PWR button briefly to cycle power levels: 25 mW → 1 W → 2 W → 3 W.

The green LED indicates the selected level by blinking:

- 25 mW: blinks once
- 1 W: blinks twice
- 2 W: blinks three times
- 3 W: blinks four times

See the diagram below for details.



Long-press PWR for 3s to enter Pit mode; the green LED turns solid on.

Short-press PWR to cycle power levels					
	Pit Mode	25mW	1W	2W	3W
Green LED	solid on (Long-press PWR for 3s)	Blink once	Blinks twice	Blinks Three times	Blinks four times

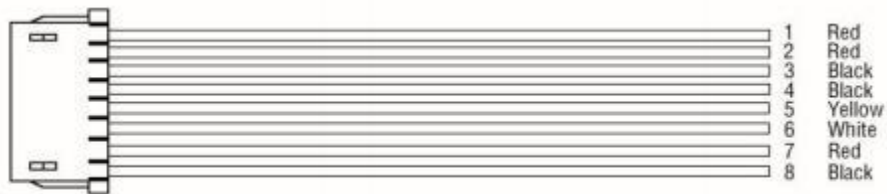
Note: This video transmitter (VTX) is equipped with thermal protection.

If the internal temperature exceeds 100 °C, output power is reduced by one step; if the temperature remains above 100 °C, it is reduced again, repeating until the minimum level (1 W) is reached. Once the temperature drops to 95 °C or below, the VTX automatically restores the originally selected power level.

2.2 Channel-to-Frequency Table

Channel-to-Frequency Table								
Band	CH1	CH2	CH3	CH4	CH5	CH6	CH7	CH8
Band A	4950	4970	4990	5010	5030	5050	5070	5090
Band B	5110	5130	5150	5170	5190	5210	5230	5250
Band E	5270	5290	5310	5330	5350	5370	5390	5410
Band F	5430	5450	5470	5490	5510	5530	5550	5570
Band R	5590	5610	5630	5650	5670	5690	5710	5730
Band P	5750	5770	5790	5810	5830	5850	5870	5890
Band H	5910	5930	5950	5970	5990	6010	6030	6050
Band U	4975	5115	5255	5395	5535	5675	5815	5955

2.3 SH1.0-8P Pinout Description



Pinout Description

NO	1	2	3	4	5	6	7	8
FUN	18-36V		GND		Video	IRC	5V Out	GND
Color	Red		Black		Yellow	White	Red	Black

3 Precautions

3.1 When installing the VTX, leave adequate space around the module to guarantee air-flow and proper heat dissipation. Otherwise the over-temperature protection will activate, reducing—or even shutting off—transmit power.

3.2 Before powering on, verify that the supply voltage is within the specified range and that polarity is correct to prevent damage to the unit.

3.3 Always connect an antenna to the RF output before applying power; this extends the module's service life.

3.4 Read this manual before use to ensure correct wiring and prolong the life of the module.

3.5 Do not bend the antenna cable or power leads sharply; improper handling can degrade performance or cause failure.

3.6 Understand and comply with all local laws and regulations before operating.

3.7 This product is not intended for use by children.